TEXORA · LECTURE SHEET

Physical Structure of Fibres

The architecture inside a fibre: polymer chains and degree of polymerisation, the two-phase model of crystalline and amorphous regions, orientation, and the structural hierarchy from molecule to microfibril to fibre — the foundation on which every other topic in textile physics stands.

TE-3209 · TEXTILE PHYSICS · TOPIC 1 · LEVEL 3 TERM 2

1. The polymer chain — length matters

Every textile fibre is an assembly of long-chain molecules (polymers) — cellulose in cotton and viscose, keratin protein in wool, polyamide in nylon, polyester in PET. Chain length is counted by the **degree of polymerisation** (DP): repeat units per molecule. Native cotton cellulose runs to DP ≈ 10,000+ in the plant and roughly 3,000–5,000 in processed fibre after purification damage; regenerated viscose is deliberately cut to DP ≈ 250–450 for spinning; PET and nylon are built to modest DPs of roughly 100–200 — their strength comes from crystallinity and orientation rather than sheer chain length. DP is strength in reserve: every chain scission (bleaching damage, acid tendering, photodegradation) spends it, and the fibre's tenacity falls in proportion long before its appearance changes.

2. The two-phase model: crystalline and amorphous

Chains inside a fibre do not order themselves uniformly. In **crystalline regions** (crystallites) chain segments pack in a regular lattice, held by hydrogen bonds (cellulose, proteins) or van der Waals forces (PET, polyolefins); X-ray diffraction reads these regions as sharp spots and arcs. In **amorphous regions** chains wander disordered — accessible, flexible, thirsty. The working picture (the fringed-micelle/fringed-fibril model) has single chains threading through several crystallites, stitching the structure together: remove that stitching and the fibre would fall apart into powder.

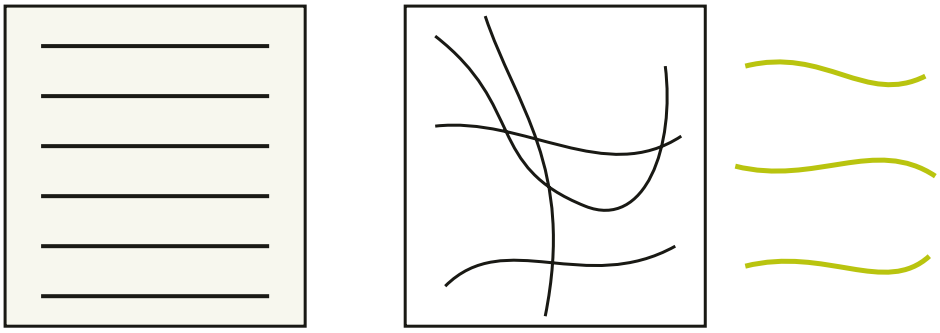


Figure 1 — The two-phase structure: ordered crystallites and disordered amorphous regions, with single molecules passing through both and binding the fibre together.

Typical crystallinity (X-ray/density estimates, method-dependent): cotton 65–70%, viscose ≈ 30–40%, wool ≈ 25–30% (its order is of a different, helical kind), drawn PET ≈ 50–60%, nylon ≈ 40–50%. The proportions explain the fibre personalities encountered through this course: crystalline regions give strength, stiffness and chemical resistance but exclude water and dye; amorphous regions admit moisture and dye, provide extension and flexibility — every property in Topics 3–10 is this balance, read from a different angle.

WORKED EXAMPLE — CRYSTALLINITY FROM DENSITY

Density separates the phases because crystals pack tighter. Take cellulose: crystal density ρ_c = 1.59 g/cm³, amorphous density ρ_a = 1.25 g/cm³, and a fibre sample of ρ = 1.45 g/cm³. Volume-fraction crystallinity = $(\rho - \rho_a)/(\rho_c - \rho_a)$ = (1.45 − 1.25)/(1.59 − 1.25) = 0.20/0.34 = **0.59 ≈ 59%**. (Convert with care: mass-fraction and volume-fraction definitions differ slightly; published values always carry their method with them.) The measurement is quick, cheap and destructive of nothing but a little sample — the standard first estimate before X-ray confirmation.

3. Orientation

Chains and crystallites also have a **direction**: in a well-drawn synthetic filament they align along the fibre axis; in cotton they spiral in microfibrils around the lumen; in an undrawn filament they are nearly random. Orientation is created by drawing (the draw ratio of melt spinning) and by growth (cotton's secondary wall). Its fingerprints are everywhere in this course: birefringence (Topic 6) measures it optically, tenacity rises with it, elongation falls, and dye uptake — set by the amorphous fraction and its accessibility — falls as drawing squeezes the structure tight. Natural fibres arrive with their orientation grown in; synthetics have theirs engineered, which is why one polymer (PET) can be sold as textile filament, high-tenacity tyre yarn or staple for blending, differentiated almost purely by draw ratio.

4. The structural hierarchy

LEVEL	WHAT IT IS — AND THE TEXTILE EXAMPLE
Repeat unit / chain	Glucose units in cellulose (β-1,4 links), amino acids in keratin, ester units in PET; chain conformation (helix in wool α-keratin) sets elasticity
Crystallite / microfibril	Cotton microfibrils wind helically in the secondary wall (fibrillar angle ≈ 20–30° to the axis) — the spiral explains cotton's convolutions and its strength gain when wet
Macro-structure	Cotton: cuticle, primary wall, secondary wall, lumen. Wool: scaly cuticle, cortex of spindle cells (ortho/para in fine wools — the source of crimp), medulla in coarse fibres
Fibre geometry	Length, fineness (diameter or linear density), cross-section shape (kidney-bean cotton, trilobal nylon for luster control), surface texture (wool scales → felting, Topic 5)

Wool's bilateral cortex deserves its sentence: ortho- and para-cortex absorb water and swell unequally, so the fibre curls — crimp is structure, not accident, and the same asymmetry drives felting. Structure, at every level, is the property list written in advance.

5. Key points

- Fibres are long-chain polymers characterised by DP: sky-high in cellulose, modest in synthetics, and consumed by chemical damage.
- Two-phase model: crystalline regions (strength, resistance, exclusion) woven with amorphous regions (access, flexibility, dye/moisture uptake) by chains threading both.
- Density gives a quick crystallinity estimate — 59% for the cellulose example.
- Orientation (grown in naturals, drawn into synthetics) trades tenacity against elongation and dyeability.
- Hierarchy matters: chain → crystallite/microfibril → wall layers → fibre geometry; cotton's spiral microfibrils and wool's bilateral cortex explain their signature behaviours.

6. Practice

Answer in your own words; brief model answers follow.

1. A polymer has repeat-unit molar mass 162 g/mol (cellulose) and number-average molar mass 486,000 g/mol. Find its DP.

2. Using the density method with $\rho_c = 1.59$ and $\rho_a = 1.25$ g/cm³, find the volume-fraction crystallinity of a regenerated cellulose film of density 1.33 g/cm³, and comment against cotton's ≈ 65–70%.
3. Why does a chain that passes through several crystallites matter mechanically — what would a fibre of separate, unconnected crystallites be like?
4. Two PET filaments have the same polymer and DP; one is drawn 5×, the other 2×. Predict the ranking of tenacity, breaking extension and dye uptake, with reasons.
5. Explain why dyeing and moisture absorption are said to happen "in the amorphous regions only".
6. Cotton gains strength when wet while viscose loses heavily. Frame both facts in two-phase structure terms (developed further in Topic 4).

Answer key

1. DP = 486,000/162 = 3,000 — typical processed cotton cellulose.
2. (1.33 − 1.25)/(1.59 − 1.25) = 0.08/0.34 = 0.235 ≈ 24% — far below cotton: regeneration dissolves the native order and the re-formed fibre crystallises only partially, hence viscose's lower strength and higher uptake.
3. The threading chains (tie molecules) transmit load between crystallites and stop them sliding apart; without them the assembly is a powder of crystals with no tensile integrity — strength lives in the connections, not the crystals alone.
4. Drawn 5×: higher tenacity (chains aligned to the load), lower extension (alignment already spent), lower dye uptake (tighter, more crystalline packing excludes dye). Drawn 2× reverses all three.
5. Crystallites are densely packed lattices with no room for guest molecules; water and dye can only enter the disordered regions, swelling them apart — hence uptake scales with amorphous fraction and its accessibility.
6. Cotton: water enters amorphous regions and lets chains settle into more uniform load-sharing (better hydrogen-bond registry on swelling), so wet strength rises. Viscose: its crystallinity is low and its amorphous regions carry most of the load-bearing connections; plasticising them with water cuts wet strength by roughly half. Same model, opposite ends of the crystallinity scale.
